

Phase 6: DOM Manipulation Practice Sheet

Easy Level (Q1 - Q4)

These questions focus on selecting elements and modifying the DOM.

Question 1 — Change Heading Text

HTML

```
<h1 id="title">Welcome</h1>
```

Using JavaScript, change the heading text to:

```
Welcome to JavaScript DOM
```

Expected Output

The webpage should display:

```
Welcome to JavaScript DOM
```

Scenario

A website wants to change the heading after the page loads.

What is this question asking?

Select an element using its **id** and update its text.

Concepts Tested

- document.getElementById()
 - textContent
-

Question 2 — Change Background Color

HTML

```
<button id="btn">Change Color</button>
```

When the button is clicked, change the page background color to **lightblue**.

Expected Output

Clicking the button changes the background color.

Scenario

Implement a simple theme switcher.

What is this question asking?

Listen for a button click and modify a CSS property.

Concepts Tested

- querySelector()
 - addEventListener()
 - style
-

Question 3 — Display Input Value

HTML

```
<input type="text" id="username">  
<button>Show</button>  
<p id="output"></p>
```

When the button is clicked, display the input value inside the paragraph.

Example

Input:

Ritik

Output:

Ritik

Scenario

A registration form displays the entered username.

Concepts Tested

- value
- textContent
- DOM Selection

Question 4 — Hide and Show Text

HTML

```
<p id="message">Hello World</p>  
<button>Toggle</button>
```

Clicking the button should hide the paragraph.

Clicking again should show it.

Scenario

A "Read More" section on a website.

Concepts Tested

- `style.display`
- `addEventListener()`

Moderate Level (Q5 - Q8)

Question 5 — Add New List Item

HTML

```
<ul id="list">
  <li>Apple</li>
</ul>

<button>Add Fruit</button>
```

When the button is clicked, add:

Banana

to the list.

Expected Output

Apple
Banana

Scenario

Adding tasks in a Todo App.

What is this question asking?

Create a new element and insert it into the DOM.

Concepts Tested

- createElement()
 - appendChild()
-

Question 6 — Remove a Card

HTML

```
<div class="card">
  <h2>JavaScript</h2>
  <button>Delete</button>
</div>
```

Clicking the button should remove the card completely.

Scenario

Deleting a note in a Notes App.

Concepts Tested

- remove()
 - parentElement
-

Question 7 — Counter App

HTML

```
<h1>0</h1>

<button>+</button>

<button>-</button>
```

Requirements:

- Increase count
- Decrease count
- Display updated value

Expected Output

```
0  
1  
2  
1  
0
```

Scenario

Shopping cart quantity selector.

Concepts Tested

- Events
- Variables
- DOM Updates

Question 8 — Live Character Counter

HTML

```
<textarea></textarea>  
  
<p>Characters: 0</p>
```

As the user types, update the character count automatically.

Example

Typing:

Hello

should display:

Characters: 5

Scenario

Twitter/X post character counter.

Concepts Tested

- input event
- value.length

Hard Level (Q9 - Q12)

Question 9 — Todo List

Build a Todo App.

HTML

```
<input>

<button>Add</button>

<ul></ul>
```

Requirements:

- Add new task
- Display task
- Clear input after adding

Example

Input:

```
Learn JavaScript
```

Output:

```
• Learn JavaScript
```

Scenario

Task management application.

What is this question asking?

Combine user input, DOM creation, and dynamic rendering.

Concepts Tested

- createElement()
- appendChild()
- value
- Events

Question 10 — Image Gallery

HTML

```
<img id="image">  
  
<button>Next</button>
```

Store multiple image URLs in an array.

Every click should show the next image.

Scenario

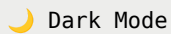
Portfolio image slider.

Concepts Tested

- Arrays
- DOM
- src attribute
- Events

Question 11 — Dark Mode Toggle

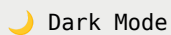
Create a button:

A button with a yellow moon icon and the text "Dark Mode".

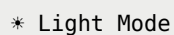
Requirements:

- Switch between Light and Dark Mode.
- Change background color.
- Change text color.
- Change button text.

Example

A button with a yellow moon icon and the text "Dark Mode".

↓

A button with a yellow star icon and the text "Light Mode".

Scenario

Modern websites like YouTube and GitHub.

What is this question asking?

Maintain application state while updating multiple DOM elements.

Concepts Tested

- `classList.toggle()`
- `style`
- `textContent`

Question 12 — Mini Student Manager (Interview Level)

Each student has:

```
{  
  name,  
  course  
}
```

HTML

```
<input placeholder="Name">  
  
<input placeholder="Course">  
  
<button>Add Student</button>  
  
<div id="students"></div>
```

Requirements:

- Add student
- Display student cards
- Delete any student

- Show total students

Example

```
Name: Ritik  
Course: MERN
```



```
-----  
Ritik  
MERN  
Delete  
-----  
  
Total Students: 1
```

Scenario

Student Management Dashboard.

What is this question asking?

Build a small CRUD application using DOM manipulation.

Concepts Tested

- DOM Selection
- createElement()
- appendChild()
- remove()
- Arrays
- Objects
- Event Handling
- Dynamic Rendering